

CS182 HW0

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Problem 2

(a) Show $\frac{\partial}{\partial x}(x^T c) = c^T$.

Let $f(x) = x^T c = \sum_{i=1}^n x_i c_i$. Then

$$\frac{\partial f}{\partial x_j} = c_j \quad \text{for each } j = 1, \dots, n.$$

Hence,

$$\frac{\partial f}{\partial x} = [c_1, c_2, \dots, c_n] = c^T.$$

(b) Show $\frac{\partial}{\partial x} \|x\|_2^2 = 2x^T$.

Let $f(x) = \|x\|_2^2 = x^T x = \sum_{i=1}^n x_i^2$. Then

$$\frac{\partial f}{\partial x_j} = 2x_j,$$

so the full gradient is

$$\frac{\partial f}{\partial x} = [2x_1, 2x_2, \dots, 2x_n] = 2x^T.$$

(c) Show $\frac{\partial}{\partial x}(Ax) = A$.

Let $f(x) = Ax$, where $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$. The Jacobian of f is the $n \times n$ matrix whose (i, j) -th entry is

$$\frac{\partial}{\partial x_j}(Ax)_i = \frac{\partial}{\partial x_j} \left(\sum_{k=1}^n A_{ik} x_k \right) = A_{ij}.$$

So the Jacobian is A .

(d) Show $\frac{\partial}{\partial x}(x^T Ax) = x^T(A + A^T)$.

Let $f(x) = x^T Ax = \sum_{i=1}^n \sum_{j=1}^n x_i A_{ij} x_j$. Then:

$$\frac{\partial f}{\partial x_k} = \sum_{j=1}^n A_{kj} x_j + \sum_{i=1}^n x_i A_{ik} = (Ax)_k + (A^T x)_k.$$

Therefore,

$$\frac{\partial f}{\partial x} = x^T A + x^T A^T = x^T (A + A^T).$$

(e) Under what condition is the previous derivative equal to $2x^T A$?

We have

$$x^T (A + A^T) = 2x^T A \iff A + A^T = 2A \iff A = A^T.$$

So the previous derivative equals $2x^T A$ if and only if A is symmetric.

Problem 3

a) We want to find the vector w that minimizes the squared Euclidean norm of the residual, $\|Xw - y\|^2$. Let's define the loss function $L(w)$:

$$\begin{aligned} L(w) &= \|Xw - y\|^2 \\ &= (Xw - y)^T (Xw - y) \\ &= (w^T X^T - y^T) (Xw - y) \\ &= w^T X^T Xw - w^T X^T y - y^T Xw + y^T y \end{aligned}$$

Since $y^T Xw$ is a scalar, it is equal to its transpose $(y^T Xw)^T = w^T X^T y$. So,

$$L(w) = w^T X^T Xw - 2w^T X^T y + y^T y$$

To find the minimum, we take the gradient of $L(w)$ with respect to w and set it to zero.

$$\nabla_w L(w) = 2X^T Xw - 2X^T y$$

Setting the gradient to zero:

$$\begin{aligned} 2X^T Xw - 2X^T y &= 0 \\ X^T Xw &= X^T y \end{aligned}$$

This is the normal equation. Assuming $X^T X$ is invertible (which it is when $m > n$ and X has full column rank), the least-squares solution w^* is:

$$w^* = (X^T X)^{-1} X^T y$$

b) We substitute $X = U\Sigma V^T$ into the normal equation solution. First, we find the terms $X^T X$ and its inverse.

$$X^T = (U\Sigma V^T)^T = V\Sigma^T U^T$$

Then,

$$X^T X = (V\Sigma^T U^T)(U\Sigma V^T)$$

Since U is an orthonormal matrix, $U^T U = I$.

$$X^T X = V \Sigma^T \Sigma V^T$$

Now, we find the inverse. Since V is orthonormal, $(V^T)^{-1} = V$ and $V^{-1} = V^T$.

$$(X^T X)^{-1} = (V \Sigma^T \Sigma V^T)^{-1} = V (\Sigma^T \Sigma)^{-1} V^T$$

Now we substitute this back into the expression for w^* :

$$\begin{aligned} w^* &= \underbrace{(V (\Sigma^T \Sigma)^{-1} V^T)}_{(X^T X)^{-1}} \underbrace{(V \Sigma^T U^T)}_{X^T} y \\ &= V (\Sigma^T \Sigma)^{-1} (V^T V) \Sigma^T U^T y \end{aligned}$$

Since V is orthonormal, $V^T V = I$.

$$w^* = V (\Sigma^T \Sigma)^{-1} \Sigma^T U^T y$$

For the overdetermined case ($m > n$), Σ is an $m \times n$ matrix. Let D be the $n \times n$ diagonal matrix of singular values σ_i . Then $\Sigma = \begin{pmatrix} D \\ 0 \end{pmatrix}$. Then $\Sigma^T \Sigma = \begin{pmatrix} D & 0 \end{pmatrix} \begin{pmatrix} D \\ 0 \end{pmatrix} = D^2$, which is an invertible $n \times n$ matrix. The term $(\Sigma^T \Sigma)^{-1} \Sigma^T = (D^2)^{-1} \begin{pmatrix} D & 0 \end{pmatrix} = \begin{pmatrix} D^{-1} & 0 \end{pmatrix}$.

$$w^* = V \Sigma^\dagger U^T y$$

c) The least-squares solution is in the form $w^* = Ay$, where $A = V \Sigma^\dagger U^T$ and A is the Moore-Penrose pseudoinverse of X . We left-multiply X by our matrix A :

$$\begin{aligned} AX &= (V \Sigma^\dagger U^T)(U \Sigma V^T) \\ &= V \Sigma^\dagger (U^T U) \Sigma V^T \end{aligned}$$

Since $U^T U = I_m$, we have:

$$AX = V (\Sigma^\dagger \Sigma) V^T$$

For the $m > n$ case, $\Sigma^\dagger = \begin{pmatrix} D^{-1} & 0 \end{pmatrix}$ and $\Sigma = \begin{pmatrix} D \\ 0 \end{pmatrix}$. Their product is:

$$\Sigma^\dagger \Sigma = \begin{pmatrix} D^{-1} & 0 \end{pmatrix} \begin{pmatrix} D \\ 0 \end{pmatrix} = D^{-1} D = I_n$$

Substituting this back:

$$AX = V I_n V^T = V V^T$$

Since V is an $n \times n$ orthonormal matrix, $V V^T = I_n$. Therefore,

$$AX = I_n$$

Since $AX = I$, the matrix A acts as a left-inverse for X .

d) For the case $m < n$, the system $Xw = y$ is underdetermined and has infinitely many solutions. We want the solution with the minimum Euclidean norm. We use Lagrange multipliers. The Lagrangian is:

$$\mathcal{L}(w, \lambda) = w^T w + \lambda^T (y - Xw)$$

Taking the gradient with respect to w and setting it to zero:

$$\nabla_w \mathcal{L} = 2w - X^T \lambda = 0 \implies w = \frac{1}{2} X^T \lambda$$

This shows that the optimal w^* must lie in the row space of X . We substitute this into $Xw = y$:

$$X \left(\frac{1}{2} X^T \lambda \right) = y \implies XX^T \lambda = 2y$$

When $m < n$ and X has full row rank, the $m \times m$ matrix XX^T is invertible. We can solve for λ :

$$\lambda = 2(XX^T)^{-1}y$$

Finally, we substitute λ back into the expression for w :

$$w^* = \frac{1}{2} X^T (2(XX^T)^{-1}y) = X^T (XX^T)^{-1}y$$

e) We substitute $X = U\Sigma V^T$ into the min-norm solution $w^* = X^T (XX^T)^{-1}y$. First, let's compute XX^T :

$$XX^T = (U\Sigma V^T)(U\Sigma V^T)^T = (U\Sigma V^T)(V\Sigma^T U^T)$$

Since $V^T V = I$, we get:

$$XX^T = U\Sigma\Sigma^T U^T$$

The inverse is:

$$(XX^T)^{-1} = (U\Sigma\Sigma^T U^T)^{-1} = U(\Sigma\Sigma^T)^{-1}U^T$$

Now substitute everything into the expression for w^* :

$$\begin{aligned} w^* &= \underbrace{(V\Sigma^T U^T)}_{X^T} \underbrace{(U(\Sigma\Sigma^T)^{-1}U^T)}_{(XX^T)^{-1}} y \\ &= V\Sigma^T (U^T U) (\Sigma\Sigma^T)^{-1} U^T y \end{aligned}$$

Since $U^T U = I$:

$$w^* = V\Sigma^T (\Sigma\Sigma^T)^{-1} U^T y$$

For the underdetermined case ($m < n$), Σ is an $m \times n$ matrix. Let D be the $m \times m$ diagonal matrix of singular values. Then $\Sigma = \begin{pmatrix} D & 0 \end{pmatrix}$. Then $\Sigma\Sigma^T = \begin{pmatrix} D & 0 \end{pmatrix} \begin{pmatrix} D \\ 0 \end{pmatrix} = D^2$, which is

an invertible $m \times m$ matrix. The term $\Sigma^T(\Sigma\Sigma^T)^{-1} = \begin{pmatrix} D \\ 0 \end{pmatrix} (D^2)^{-1} = \begin{pmatrix} D^{-1} \\ 0 \end{pmatrix}$. This is the definition of $\Sigma^\dagger$ for the fat matrix case.

$$\mathbf{w}^* = \mathbf{V}\Sigma^\dagger\mathbf{U}^T\mathbf{y}$$

f) The min-norm solution is in the form $\mathbf{w}^* = \mathbf{B}\mathbf{y}$, where $\mathbf{B} = \mathbf{V}\Sigma^\dagger\mathbf{U}^T = \mathbf{X}^\dagger$. Let's right-multiply $\mathbf{X}$ by our matrix $\mathbf{B}$:

$$\begin{aligned} \mathbf{XB} &= (\mathbf{U}\Sigma\mathbf{V}^T)(\mathbf{V}\Sigma^\dagger\mathbf{U}^T) \\ &= \mathbf{U}\Sigma(\mathbf{V}^T\mathbf{V})\Sigma^\dagger\mathbf{U}^T \end{aligned}$$

Since $\mathbf{V}^T\mathbf{V} = \mathbf{I}_n$, we have:

$$\mathbf{XB} = \mathbf{U}(\Sigma\Sigma^\dagger)\mathbf{U}^T$$

For the $m < n$ case, $\Sigma = \begin{pmatrix} D & 0 \end{pmatrix}$ and $\Sigma^\dagger = \begin{pmatrix} D^{-1} \\ 0 \end{pmatrix}$. Their product is:

$$\Sigma\Sigma^\dagger = \begin{pmatrix} D & 0 \end{pmatrix} \begin{pmatrix} D^{-1} \\ 0 \end{pmatrix} = DD^{-1} = \mathbf{I}_m$$

Substituting this back:

$$\mathbf{XB} = \mathbf{U}\mathbf{I}_m\mathbf{U}^T = \mathbf{U}\mathbf{U}^T$$

Since $\mathbf{U}$ is an $m \times m$ orthonormal matrix, $\mathbf{U}\mathbf{U}^T = \mathbf{I}_m$. Therefore,

$$\mathbf{XB} = \mathbf{I}_m$$

Since $\mathbf{XB} = \mathbf{I}$, the matrix $\mathbf{B}$ acts as a right-inverse for $\mathbf{X}$. $\mathbf{T}$

Problem 4

a) We want to minimize this:

$$\arg \min_{\mathbf{w}} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \lambda \|\mathbf{w}\|_2^2$$

Expanding,

$$\begin{aligned} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \lambda \|\mathbf{w}\|_2^2 &= (\mathbf{y} - \mathbf{X}\mathbf{w})^T(\mathbf{y} - \mathbf{X}\mathbf{w}) + \lambda \mathbf{w}^T\mathbf{w} \\ &= \mathbf{y}^T\mathbf{y} - 2\mathbf{y}^T\mathbf{X}\mathbf{w} + \mathbf{w}^T\mathbf{X}^T\mathbf{X}\mathbf{w} + \lambda \mathbf{w}^T\mathbf{w} \\ &= \mathbf{y}^T\mathbf{y} - 2\mathbf{w}^T\mathbf{X}^T\mathbf{y} + \mathbf{w}^T(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})\mathbf{w} \end{aligned}$$

Taking gradient with respect to $\mathbf{w}$ and setting it to zero:

$$\nabla_{\mathbf{w}} = -2\mathbf{X}^T\mathbf{y} + 2(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})\mathbf{w} = 0 \Rightarrow (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})\mathbf{w} = \mathbf{X}^T\mathbf{y} \Rightarrow \boxed{\mathbf{w} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}}$$

b) Let $X = U\Sigma V^T$ be the SVD of X . Then:

$$X^T X = V\Sigma^T \Sigma V^T \quad \text{so} \quad X^T X + \lambda I = V(\Sigma^T \Sigma + \lambda I)V^T$$

Hence,

$$(X^T X + \lambda I)^{-1} X^T = V(\Sigma^T \Sigma + \lambda I)^{-1} \Sigma^T U^T$$

This means the singular values σ_i are replaced with:

$$\frac{\sigma_i}{\sigma_i^2 + \lambda}$$

So, when $\sigma_i \ll \lambda$: $\frac{\sigma_i}{\sigma_i^2 + \lambda} \approx \frac{\sigma_i}{\lambda} \rightarrow 0$. Also, when $\sigma_i \gg \lambda$: $\frac{\sigma_i}{\sigma_i^2 + \lambda} \approx \frac{1}{\sigma_i}$, same as pseudoinverse.

c) We assume the prior $\mathbf{w} \sim \mathcal{N}(0, I)$, and $Y = X\mathbf{w} + \sqrt{\lambda}N$ where $N \sim \mathcal{N}(0, I)$. Then $Y|\mathbf{w} \sim \mathcal{N}(X\mathbf{w}, \lambda I)$.

Using Bayes' rule:

$$\text{MAP estimate} = \arg \max_{\mathbf{w}} \log p(Y|\mathbf{w}) + \log p(\mathbf{w})$$

This becomes minimizing:

$$\frac{1}{\lambda} \|Y - X\mathbf{w}\|^2 + \|\mathbf{w}\|^2 \Rightarrow \|Y - X\mathbf{w}\|^2 + \lambda \|\mathbf{w}\|^2$$

So MAP estimate is:

$$\boxed{\mathbf{w} = (X^T X + \lambda I)^{-1} X^T Y}$$

d) ??

e) Now define:

$$\tilde{X} = [X \quad \sqrt{\lambda} I_n]$$

Let $\boldsymbol{\eta} \in \mathbb{R}^{d+n}$ with:

$$\tilde{X}\boldsymbol{\eta} = \mathbf{y}, \quad \text{and we minimize } \|\boldsymbol{\eta}\|^2$$

Then:

$$\min_{\boldsymbol{\eta}} \|\boldsymbol{\eta}\|^2 \quad \text{s.t.} \quad \tilde{X}\boldsymbol{\eta} = \mathbf{y} \Rightarrow \text{min-norm solution to underdetermined system}$$

The first d coordinates of $\boldsymbol{\eta}$ equal:

$$\boxed{\mathbf{w} = (X^T X + \lambda I)^{-1} X^T \mathbf{y}}$$

f) ??

g) We have:

$$\mathbf{w}_\lambda = (X^T X + \lambda I)^{-1} X^T \mathbf{y}$$

As $\lambda \rightarrow \infty$:

$$(X^T X + \lambda I)^{-1} \approx \frac{1}{\lambda} I \Rightarrow \mathbf{w}_\lambda \rightarrow 0$$

The weights are pushed toward zero.

h) If X is tall (full column rank), then:

$$\lim_{\lambda \rightarrow 0} \mathbf{w}_\lambda = (X^T X)^{-1} X^T \mathbf{y} \Rightarrow \text{Ordinary Least squares solution}$$

If X is underdetermined, then $X^T X$ is not invertible. But ridge adds regularization so we get a unique solution. As $\lambda \rightarrow 0$, we approach the minimum solution.

Problem 5

a)

(i) The elbow location e is where the ReLU switches from 0 to positive:

$$wx + b = 0 \quad \Rightarrow \quad e = -\frac{b}{w} \quad (\text{if } w \neq 0)$$

(ii) Derivative of the loss w.r.t. $\phi(x)$:

$$\frac{d\ell}{d\phi} = \phi(x) - y$$

(iii) Partial derivative w.r.t. w :

$$\frac{\partial \ell}{\partial w} = \frac{d\ell}{d\phi} \cdot \frac{d\phi}{dw} = \begin{cases} (\phi(x) - y) \cdot x & \text{if } wx + b > 0 \\ 0 & \text{otherwise} \end{cases}$$

(iv) Partial derivative w.r.t. b :

$$\frac{\partial \ell}{\partial b} = \frac{d\ell}{d\phi} \cdot \frac{d\phi}{db} = \begin{cases} (\phi(x) - y) & \text{if } wx + b > 0 \\ 0 & \text{otherwise} \end{cases}$$

b)

(i) If $\phi(x) = 0$: the ReLU is inactive, so no gradient flows. No update.

(ii) If $w > 0$, $x > 0$, and $\phi(x) > 0$:

$$\frac{\partial \ell}{\partial w} = x, \quad \frac{\partial \ell}{\partial b} = 1$$

SGD update with learning rate λ :

$$w' = w - \lambda x, \quad b' = b - \lambda$$

New elbow location:

$$e' = -\frac{b - \lambda}{w - \lambda x}$$

(iii) If $w > 0$, $x < 0$, and $\phi(x) > 0$:

$$\frac{\partial \ell}{\partial w} = x < 0, \quad \Rightarrow w' = w - \lambda x \text{ (increases)}$$

$$b' = b - \lambda \Rightarrow \text{elbow moves left}$$

(iv) If $w < 0$, $x > 0$, and $\phi(x) > 0$:

$$\frac{\partial \ell}{\partial w} = x > 0, \quad w' = w - \lambda x \text{ (more negative)}$$

$$b' = b - \lambda \Rightarrow \text{elbow moves right}$$

c) Each ReLU $\phi_i(x) = \max(0, w_i x + b_i)$. The elbow of the i th ReLU is:

$$e_i = -\frac{b_i}{w_i}$$

d) Gradient descent:

$$w'_i = w_i - \lambda \frac{\partial \ell}{\partial w_i}, \quad b'_i = b_i - \lambda \frac{\partial \ell}{\partial b_i}$$

If $\phi_i(x)$ is active, we get:

$$\frac{\partial \ell}{\partial w_i} = (\phi_i(x) - y)x, \quad \frac{\partial \ell}{\partial b_i} = (\phi_i(x) - y)$$

Then:

$$e'_i = -\frac{b'_i}{w'_i} = -\frac{b_i - \lambda(\phi_i(x) - y)}{w_i - \lambda(\phi_i(x) - y)x}$$